Live Steps: TL;DR

Rules for every step

- Work **top to bottom**. Delete the `//TODO` line along with the line it names. Each line number is where the TODO sits when you reach that row, with the rows above it already done.
- `TalkDemo.cs` always goes last, then run the option. Each option shows **Bad** first. Press a key to show **Good**.
- **Finding the code:** every change sits under a comment that starts `//TODO: Talk Step N`. Press `Ctrl+Shift+F`, set the scope to Entire Solution, and search for the text in the Search row of that step.
- Lost? Use the Visual Studio Task List and filter it by `Talk Step N`.

Slide	File	Line	Change	Problem this fixes (slide)	Option
1–5	Intro, no code				
6–7	**Step 1 problem:** any caller can build any order. `new Order()` is empty, and `Status = "Banana"`, `Total = -99.99` saves and reloads through EF Core.				
8–10	Fix: private constructor, `Order.Place`, `Place` guards				
11	**Search:** `TODO: Talk Step 1` (3 TODOs in 2 files)				
	`OrderModels/Domain/Order.cs`	36	`public Order()` → `private Order()`	Any caller can write `new Order { Status = "Banana", Total = -99.99m }`. (6, 7)	
	`OrderModels/Domain/Order.cs`	40	`Place`: uncomment 2 guards	An order can have a blank customer and zero lines. (6, 7)	
	`TalkApp/TalkDemo.cs`	138	`Solution1`: delete throw, uncomment code	Bad: the Banana order saves and reloads. Good: no public constructor, and `Place` rejects a blank customer and no lines.	**1**
12	**Step 2 problem:** every setter is public, so any caller can rewrite a placed order.				
13	Fix: private setters. Slide 13 shows five properties; `Lines` gets one too.				
14	**Search:** `TODO: Talk Step 2` (7 TODOs in 2 files)				
	`OrderModels/Domain/Order.cs`	5	`Id` → `private set`	Any caller can change `Id`. (12)	
	`OrderModels/Domain/Order.cs`	7	`Status` → `private set`	Any caller can set `Status` to anything. (12, 15)	
	`OrderModels/Domain/Order.cs`	9	`Total` → `private set`	`order.Total = 0.01m` works on a placed order. (12)	
	`OrderModels/Domain/Order.cs`	14	`OrderDate` → `private set`	`order.OrderDate = DateTime.Now.AddDays(-30)` moves the date back. (12)	

Slide	File	Line	Change	Problem this fixes (slide)	Option
	`OrderModels/Domain/Order.cs`	16	`CustomerName` → `private set`	`order.CustomerName = "Someone Else"` gives the order to another customer. (12)	
	`OrderModels/Domain/Order.cs`	18	`Lines` → `private set`	`order.Lines = new()` replaces every line. Every setter is public. `Lines.Add` still works until Step 5. (12)	
	`TalkApp/TalkDemo.cs`	215	`Solution2`: delete throw, uncomment code	Bad: moved date, Someone Else, total 0.01. Good: all five setters are private, and `Place` still creates the order.	**2**
15	**Step 3 problem:** the anemic `Order` ships a Placed order and a Cancelled order. The domain `Order` now blocks the assignment, but nothing can move `Status` forward.				
16	Fix: `Process`, `Confirm`, `Ship`, `Cancel` guard each transition				
17	**Search:** `TODO: Talk Step 3` (5 TODOs in 2 files)				
	`OrderModels/Domain/Order.cs`	57	`Process`: delete throw, uncomment 3 lines	With a private setter, nothing can move Placed to Processing. (15)	
	`OrderModels/Domain/Order.cs`	64	`Confirm`: delete throw, uncomment 3 lines	With a private setter, nothing can move Processing to Confirmed. (15)	
	`OrderModels/Domain/Order.cs`	71	`Ship`: delete throw, uncomment 3 lines	A Placed order that was never confirmed can ship, and so can a Cancelled order. (15)	
	`OrderModels/Domain/Order.cs`	78	`Cancel`: delete throw, uncomment 3 lines	Nothing stops a shipped order from being cancelled. (16)	
	`TalkApp/TalkDemo.cs`	259	`Solution3`: delete throw, uncomment code	Bad: Placed and Cancelled orders both print Shipped. Good: `Ship` throws for both, and `Process`, `Confirm`, `Ship` reaches Shipped.	**3**
18	**Step 4 problem:** a decimal `Total` has no currency. 19.99 USD plus 25.00 EUR gives 44.99. Dollars? Euros?				
19	**Step 4 problem:** `Money` checks nothing. `Money.Create` accepts -42.50, and USD plus EUR returns 44.99 USD.				
20–22	Fix: `Money` total, `OwnsOne`, `Money` guards				
23	**Search:** `TODO: Talk Step 4` (6 TODOs in 4 files)				
	`OrderModels/Domain/Money.cs`	16	`Create`: uncomment 2 guards	`Money.Create(-42.50m, "USD")` is accepted, and so is a blank currency. (19, 22)	
	`OrderModels/Domain/Money.cs`	24	`Add`: uncomment currency guard	19.99 USD plus 25.00 EUR returns 44.99 USD. (19)	
	`OrderModels/Domain/Order.cs`	11	Delete `decimal Total` above the TODO, uncomment `Money Total`	A decimal `Total` has no currency. (18)	

Slide	File	Line	Change	Problem this fixes (slide)	Option
	`OrderModels/Domain/Order.cs`	83	`RecalculateTotal`: `Sum` → `Aggregate(... Add)`	`Sum` adds bare decimals, so USD and EUR mix without an error. (18)	
	`TalkApp/Data/TalkDbContext.cs`	42	`HasPrecision` → `OwnsOne` (uncomment 5 lines)	The `Total` column in SQL Server holds a bare number with no currency. (18, 21)	
	`TalkApp/TalkDemo.cs`	321	`Solution4`: delete throw, uncomment code	Bad: 44.99 with no currency, -42.50 accepted, USD plus EUR accepted. Good: 59.97 USD reloads, and the bad cases are rejected.	**4**
24	**Step 5 problem:** any caller can change `Lines`, and nothing recalculates `Total`. Lines sum to 20.00 with a total of 10.00. Zero lines still shows 10.00.				
25–26	Fix: `_lines` backing field, read-only `Lines`, `AddLine`, internal constructor, `HasField`				
27	**Step 5 problem:** nothing checks the lines. Quantity -3, product 0, and product 2001 twice are all accepted.				
28	Fix: guards in `AddLine` and the `OrderLine` constructor				
29	**Search:** `TODO: Talk Step 5` (8 TODOs in 4 files)				
	`OrderModels/Domain/Order.cs`	18	Delete `List<OrderLine> Lines` above the TODO, uncomment `_lines` + `IReadOnlyCollection` (2 lines)	Any caller can call `Lines.Add` or `Lines.Clear`. (24)	
	`OrderModels/Domain/Order.cs`	31	`Place`: `Lines.Add` → `AddLine`	`Place` adds lines directly, so its lines would skip the rules in `AddLine`. (25)	
	`OrderModels/Domain/Order.cs`	40	`AddLine`: uncomment duplicate-product guard	Product 2001 can go on the same order twice. (27)	
	`OrderModels/Domain/Order.cs`	42	`AddLine`: delete throw, uncomment 2 lines	Adding a line leaves `Total` stale: lines sum to 20.00, total says 10.00. (24)	
	`OrderModels/Domain/OrderLine.cs`	16	`public` → `internal` constructor	Any caller can create a line with `new OrderLine(...)`. (24)	
	`OrderModels/Domain/OrderLine.cs`	18	Constructor: uncomment 2 guards	A quantity of -3 and a product id of 0 are accepted. (27)	
	`TalkApp/Data/TalkDbContext.cs`	50	Uncomment `HasField("_lines")`	`Lines` has no setter, and finding `_lines` by name breaks if the field is renamed. (26)	
	`TalkApp/TalkDemo.cs`	401	`Solution5`: delete throw, uncomment code	Bad: total 10.00 with lines summing to 20.00, then zero lines, and all three bad lines accepted. Good: 10.00 USD becomes 20.00 USD, 2 lines reload, and the bad lines are rejected.	**5**

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30–33	Recap table, side-by-side models, questions, close				